Linear programming: exercises

29-E01 Free variables and minimization

Convert $\texttt{min 3u+v; u−v=2; u+v≤6; v≥0; u free}$ to our standard and slack forms. Give both solution mappings, find the original optimum, and explain why simply adding $u \geq 0$ is not a general conversion.

29-E02 Complete pivots and a dual certificate

Solve $\texttt{max } 4x_1+3x_2\texttt{; } 2x_1+x_2\leq 10\texttt{; } x_1+2x_2\leq 8\texttt{; } x\geq 0$, entering x_1 first. Give both complete dictionaries, each row's ratio, and all nonbasic coefficients. Construct the dual and verify feasibility and equal objectives.

29-E03 Zero ratios, ties, and symbolic perturbation

For $x_3=-x_1\texttt{; } x_4=2-x_1-x_2\texttt{; max } x_1+x_2$, enter x_1, give the legal leaving row and zero-step dictionary, then finish solving. Separately, for $x_3=2-x_1\texttt{; } x_4=2-x_1-x_2$, give the lexicographic leaving row when RHS=$(2+\varepsilon, 2+\varepsilon^2)$ and x_1 enters.

29-E04 From Phase I through the original optimum

Use shared artificial $a \geq 0$ for $\texttt{max x; } x\leq 4\texttt{; } x\geq 2\texttt{; } x\geq 0$. Give the infeasible initial slack point, the special-pivot feasible auxiliary dictionary, the auxiliary optimum, artificial removal, original-objective restoration, and Phase II. Change $x \leq 4$ to $x \leq 1$, find auxiliary optimum, and give an independent infeasibility certificate.

29-E05 Improving rays and unbounded regions

Solve $\texttt{max } 2x_1+x_2\texttt{; } x_1-x_2\leq 1\texttt{; } x\geq 0$, entering x_1 first. Give the pivoted dictionary, a feasible x^0, and nonnegative direction r with $Ar \leq 0$, $c^T r > 0$. Prove every $x^0+\theta r, \theta \geq 0$, is feasible. Change objective to $-x_1-x_2$ and explain the finite optimum on the same unbounded region.

29-E06 Preconditions of the potential model

Graph edges are s→a:−2, a→t:3, s→t:4, and a disconnected pair u→v:−1, v→u:−1. Write the s-to-t potential LP. Why is retaining all edges infeasible without making the actual s-to-t shortest distance nonfinite? Repair the model, solve it, and assess whether adding $d \geq 0$ is equivalent.

Linear programming: exercises (continued)

29-E07 Shared capacities and integrality

Two commodities each demand 1 from s to t. Each can use two branches, each branch consisting of two capacity 1 arcs. Write the commodity LP and give fractional and integral feasible solutions. Then give a unique shared capacity 1 bottleneck with demands 3/4 each:prove individual feasibility but joint infeasibility. Distinguish fractional feasibility, existence of an integer solution, and requiring one path per commodity.

29-E08 Complementary slackness and theorem boundaries

For $\max\ 2x_1+x_2;\ x_1+x_2\le 4;\ x_1\le 2;\ x\ge 0$, check x=(2,2), dual y=(1,1), and every gap product, proving optimality. For x=(1,3), identify the nonzero complementary product producing objective gap 1. Why does $\max\ x;\ 0\le x<1$ fall outside the LP trichotomy theorem?

References and attribution

Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein. Introduction to Algorithms, 3rd Edition.

Original educational synthesis by Lin Cing. Original material is licensed under CC BY 4.0. The referenced textbook is not distributed with these notes.

https://github.com/xinhi6716/computer-science-self-study